higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

ELECTRICAL PRINCIPLES AND PRACTICE
NQF LEVEL 3

(12041003)

12 November 2019 (X-Paper)
09:00–12:00

This question paper consists of 7 pages and a formula sheet of 2 pages.

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Start each section on a NEW page.
5. Use only a BLACK or BLUE pen.
6. Write neatly and legibly.

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QUESTION 1

1.1 The longer a conductor of a particular thickness, the greater its resistance.

Apart from the length of a conductor, list THREE other factors affecting the resistance of a material. (3)

1.2 The resistance of a heating element made of nickel-chrome wire is 20 Ω. The wire has a cross-sectional area of 0,043 mm^2 and a resistivity of 1,2 micro-ohm metres.

Calculate the length of the wire used. (3)

1.3 Briefly explain the term *positive temperature coefficient of resistance*. (2)

1.4 A coil has a resistance of 20 Ω at 25 °C. The temperature coefficient of resistance at 25 °C is 0,00385 per °C.

Calculate the resistance of the coil at 65 °C, assuming it has a positive temperature co-efficient. (3)

1.5 Silicon has an atomic number of 14.

State how many electrons it has in the:

1.5.1 First shell

1.5.2 Second shell

1.5.3 Third shell

(3 × 1) (3)

1.6 What are valence electrons? (1)

[15]

QUESTION 2

2.1 Complete the following sentences by writing the missing word or words next to the question number (2.1.1–2.1.4) in the ANSWER BOOK.

2.1.1 When electric current flows through a wire producing a magnetic field around the wire, it is known as …

2.1.2 A … is an example of an electromagnet used to control a circuit.

2.1.3 The phenomenon where similar poles (like poles) repel each other and opposite poles (unlike poles) attract each other, is known as …

2.1.4 The … rule is used to identify the direction of a magnetic field if the direction of the flow of current in a conductor is known.

(4 × 1) (4)

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2.2 Explain the operation of a simple DC motor. (3)

2.3 A straight conductor 25 cm long is moved with a velocity of 3 m/s at right angles to a magnetic field having a uniform flux density of 0,12 Wb/m^2.

Calculate the following:

2.3.1 The magnitude of the emf induced in the conductor (2)

2.3.2 The force required to produce this movement if the ends of this conductor were connected to a circuit drawing 450 Ma. (2)

2.4 Explain the difference between *magnetic flux* and *magnetic flux density.*

(2 + 2) (4)

2.5 You must determine the relative direction of an induced emf using Fleming's right-hand rule.

What does each of the following represent:

2.5.1 The index finger

2.5.2 The thumb

2.5.3 The middle finger

(3 × 1) (3)

2.6 A coil with a length of 500 mm consists of 30 turns.

Calculate the magnetic field strength if the current flowing through the coil is 2 A. (2)

[20]

QUESTION 3

3.1 Apply Kirchhoff's law to determine the value of V_2 in FIGURE 1. (3)

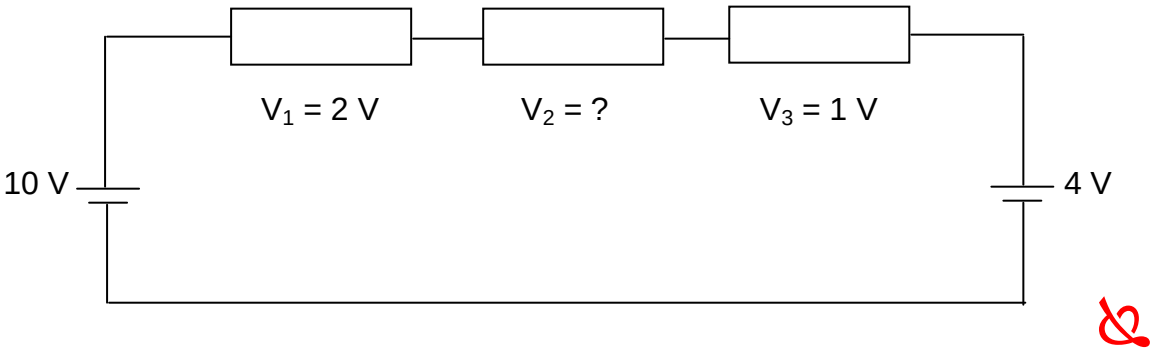


FIGURE 1

3.2 The lamps in FIGURE 2 are identical and each has a resistance of 5 Ω.

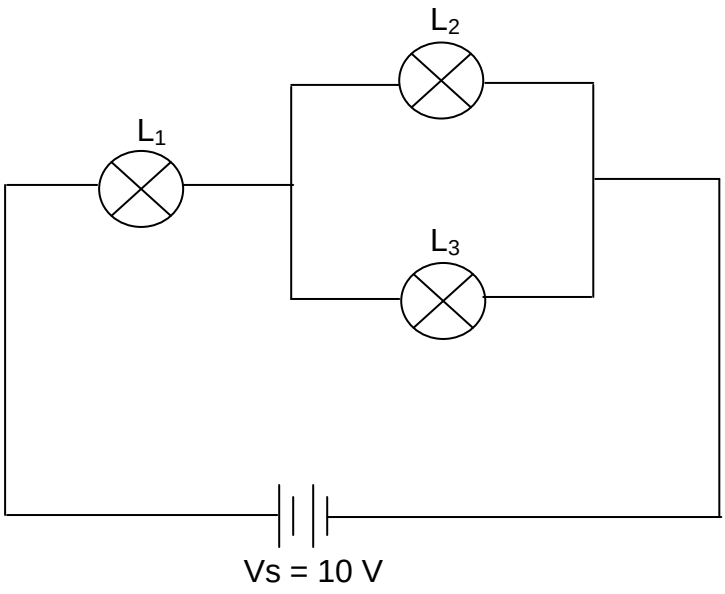


FIGURE 2

3.2.1 Determine the total resistance of the circuit. (3)

3.2.2 If lamp L2 is open circuit, calculate:

(a) The total resistance (2)
(b) The total current (2)

3.2.3 If lamp L3 is short circuit, how will it affect the resistance of lamp L1 and lamp L2? (2)

3.3 Explain the term *capacitance.* (2)

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3.4 Two capacitors of 4 µF and 2 µF are connected in parallel and this combination is then connected in series with a capacitor of 3 µF. The complete circuit is connected to a 12 V DC supply.

Calculate the following:

3.4.1 The total capacitance of the circuit (3)

3.4.2 The total charge flowing through the circuit (2)

3.4.3 The charge flowing through the 4 µF capacitor (4)

3.5 State whether the following cells are primary or secondary cells:

3.5.1 Nickel-iron cell

3.5.2 Silver-oxide cell

3.5.3 Alkaline cells

(3 × 1) (3)

3.6 Transformers play a major role in the distribution of electricity.

3.6.1 What is the function of the conservator in a transformer? (2)

3.6.2 Give TWO methods/types of dry cooling transformers. (2)
[30]

QUESTION 4

4.1 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (4.1.1–4.1.5) in the ANSWER BOOK.

4.1.1 A dynamometer-type instrument has a linear scale.

4.1.2 A moving-iron instrument is expensive.

4.1.3 A moving-coil permanent magnet-type instrument cannot be used to measure AC quantities.

4.1.4 A dynamometer-type instrument has the same basic operating principle as an electrodynamic instrument.

4.1.5 Moving-iron instruments can be used to measure in low-power circuits.

(5 × 1) (5)

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4.2 Show, with the aid of fully labelled sketches how each of the following measuring instruments can be connected to a simple circuit. Include a switch, resistor (load) and supply.

4.2.1 Voltmeter

4.2.2 Ammeter

(2 × 3) (6)

4.3 You are required to measure the current drawn by a 500 W electric heating element using a digital multimeter.

List FOUR important steps to follow when taking this measurement. (4)

[15]

QUESTION 5

5.1 Explain the term *armature reaction* as applicable to DC motors. (2)

5.2 Losses in a DC machine affect the efficiency of the machine.

List FOUR types of losses that occur in a DC machine. (4)

5.3 Complete the following table by writing the answer next to the question number (5.3.1–5.3.6) in the ANSWER BOOK.

AC MOTORS	ADVANTAGES	DISADVANTAGES
Universal motor	5.3.1	5.3.2
Shaded-pole induction motor	5.3.3	5.3.4
Capacitor-start capacitor-run motor	5.3.5	5.3.6

(6 × 1) (6)

5.4 Three-phase induction motors are the most popular electrical motors used in industry.

List FOUR applications of a three-phase induction motor. (4)

5.5 A 220 V DC motor has an armature resistance of 3 Ω and an emf of 205 V.

Calculate the armature current drawn by this motor. (3)

5.6 Name the protective device used with motors to detect when one phase has failed. (1)

[20]

TOTAL: 100

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FORMULA SHEET

1. $\cos\phi = \frac{R}{Z}$

2. $P = V\ I\ \cos\phi$

3. $Q = V\ I\ \sin\phi$

4. $S = V\ I$

5. $V_1 = \left(\frac{R_1}{R_t}\right) V_T$

6. $I_1 = \left(\frac{R_2}{R_1 + R_2}\right) I_T$

7. $R_T = R_1 + R_2 + R_3 + \cdots + R_n$

8. $\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \cdots + \frac{1}{R_n}$

9. $\frac{V_1}{V_2} = \frac{N_1}{N_2} = \frac{I_2}{I_1}$

10. $V_L = \sqrt{3}\, V\ p\ h$

11. $E = V - I_a R_a$

12. $f = \frac{Np}{60}$

13. $S = \frac{N_s - N_r}{N_s}$

14. $R_{sh} = \frac{I_m R_m}{I_{sh}}$

15. $R_{se} = \frac{V_t}{I_t} - R_m$

16. $F = B\ I\ l$

17. $\tan\phi = \sqrt{3}\left(\frac{W_1 - W_2}{W_1 + W_2}\right)$

18. $E = V + I\ r$

19. $V = I\ R$

20. $R = \frac{p\ l}{A}$

21. $R_t = R_0(1 + \alpha t)$

22. $R_t = R_0[1 + \propto_0 (t - \emptyset)]$

23. $P = V\ I$

24. $E = I^2\ R\ t$

25. $\eta = \frac{P_0}{P_{in}} \times 100$

26. $X_L = 2\pi\ f\ L$

27. $X_C = \frac{1}{2\pi\ f\ C}$

28. $Z = \sqrt{R^2 + (X_L - X_C)^2}$

29. $Mmf = N\ I$

30. $H = \frac{Mmf}{l}$

31. $B = \frac{\Phi}{A}$

32. $E = B\ l\ v$

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33. $E = -L\frac{di}{dt}$

34. $A = \frac{\pi d^2}{4}$

35. $E = \frac{1}{2} C V^2$

36. $Q = C V$

37. $F = 2\times10^{-7} \times \frac{I_1 I_2}{d}$

38. $I = \frac{Emf}{R + r_T}$

39. $L = N\frac{\Delta \phi}{\Delta I}$

40. $E = V + R_a I_a$

41. $P = \sqrt{3}\, V_l I_l \cos\theta$

42. $S = \sqrt{3}\, V_l I_l$

43. $E = -N\frac{d\emptyset}{dt}$

44. $E = \frac{V}{d}$

45. $E = \frac{1}{2} L I^2$

46. $Q = It$

47. $C_T = C_1 + C_2 + C_3 + \cdots + C_n$

48. $\frac{1}{C_T} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \cdots + \frac{1}{C_n}$

49. $L_T = L_1 + L_2 + L_3 + \cdots + L_n$

50. $\frac{1}{L_T} = \frac{1}{L_1} + \frac{1}{L_2} + \frac{1}{L_3} + \cdots + \frac{1}{L_n}$

51. $Q = \sqrt{3}\, V_l I_l \sin\theta$

52. $Pf = \cos\theta$

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